

Beyond the Final Actor: Modeling the Dual Roles of Creator and Editor for Fine-Grained LLM-Generated Text Detection

超越最终演员：建模创作者与编辑的双重角色以实现细粒度 LLM 生成文本的检测

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The misuse of large language models (LLMs) requires precise detection of synthetic text. Existing works mainly follow binary or ternary classification settings, which can only distinguish pure human/LLM text or collaborative text at best. This remains insufficient for the nuanced regulation, as the LLM-polished human text and humanized LLM text often trigger different policy consequences. In this paper, we explore fine-grained LLM-generated text detection under a rigorous four-class setting. To handle such complexities, we propose RACE (Rhetorical Analysis for Creator-Editor Modeling), a fine-grained detection method that characterizes the distinct signatures of creator and editor. Specifically, RACE utilizes Rhetorical Structure Theory (RST) to construct a logic graph for the creator's foundation while extracting Elementary Discourse Unit (EDU)-level features for the editor's style. Experiments show that RACE outperforms 12 baselines in identifying fine-grained types with low false alarms, offering a policy-aligned solution for LLM regulation.

大型语言模型（LLMs）的滥用需要对合成文本进行精确检测。现有研究主要遵循二分类或三分类的设置，只能在最佳情况下区分纯人类/LLM 文本或协作文本。这对于细致的监管来说仍然不足，因为 LLM 优化的人类文本和人性化 LLM 文本常常引发不同的政策后果。在本文中，我们探索了在严格四分类设置下的细粒度 LLM 生成文本检测。为了处理这种复杂性，我们提出了 RACE（用于创作者-编辑建模的修辞分析），这是一种细粒度检测方法，能够刻画创作者和编辑的特征差异。具体而言，RACE 利用修辞结构理论（RST）为创作者的基础构建逻辑图，同时提取编辑风格的初级话语单元（EDU）级特征。实验表明，RACE 在识别细粒度类型时具有较低的误报率，优于 12 种基线方法，为 LLM 监管提供了一种符合政策的解决方案。



Figure 1: images/47bc52eca894ec420f3c148ef120d7b3ac279a1ad6a8777e028ef4d09879dcf5.jpg

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Figure 2: images/c5ca79de4778db54407877fa9da5c503e111bcc7071ff02ad1c5dc50374c5509.jpg

Project: <https://race.yang-li.cn>

项目: <https://race.yang-li.cn>



Figure 3: [images/9ce0b16f2c267d971578d160305b84ac230475688f0d3808056fbafb322b877f.jpg](#)

Venue: ACL 2026

地点: ACL 2026



Figure 4: [images/8ba380d53e4e553588f45fef5d4766e9e081189ce2c0bcf59a18269101622441.jpg](#)

1. Introduction

1. 引言

While the surge of Large Language Models (LLMs) (OpenAI, 2025b, Yang et al., 2025a) has revolutionized content creation and inspired a diverse range of downstream applications, the improper and malicious use of LLMs is also eroding the foundation of information credibility (Anwar et al., 2024, Liu et al., 2024). From the large-scale synthesis of misinformation (Hu et al., 2025) to unauthorized academic assistance (Goodier, 2025) and LLM-based identity fraud (FBI, 2025), the ease of generating high-quality synthetic text poses a severe challenge to our trust system, necessitating effective techniques to distinguish LLM-generated text from human-written text (Wu et al., 2025a, Liu et al., 2025).

尽管大型语言模型 (LLMs) (OpenAI, 2025b, Yang 等人, 2025a) 的兴起彻底革新了内容创作, 并激发了各种下游应用的多样性, 但 LLMs 的不当和恶意使用也在侵蚀信息可信度的基础 (Anwar 等人, 2024, Liu 等人, 2024)。从大规模合成虚假信息 (Hu 等人, 2025) 到未经授权的学术帮助 (Goodier, 2025) 以及基于 LLM 的身份欺诈 (FBI, 2025), 生成高质量合成文本的便捷性对我们的信任体系构成了严峻挑战, 需要有效的技术来区分 LLM 生成的文本与人类撰写的文本 (Wu 等人, 2025a, Liu 等人, 2025)。

LLM-generated text detection was primarily formulated as a binary classification task that judges whether the given text is generated by any LLM or by human (He et al., 2024, Wang et al., 2024). However, the binary setting oversimplifies real-world scenarios where text is often a product of human-LLM

LLM 生成文本的检测最初主要被表述为一个二分类任务, 用于判断给定的文本是由任何 LLM 生成还是由人类生成 (He 等, 2024, Wang 等, 2024)。然而, 二分类的设置过于简化了现实世界中的情况, 其中文本通常是人类与 LLM 共同作用的结果。

(a)

(a)

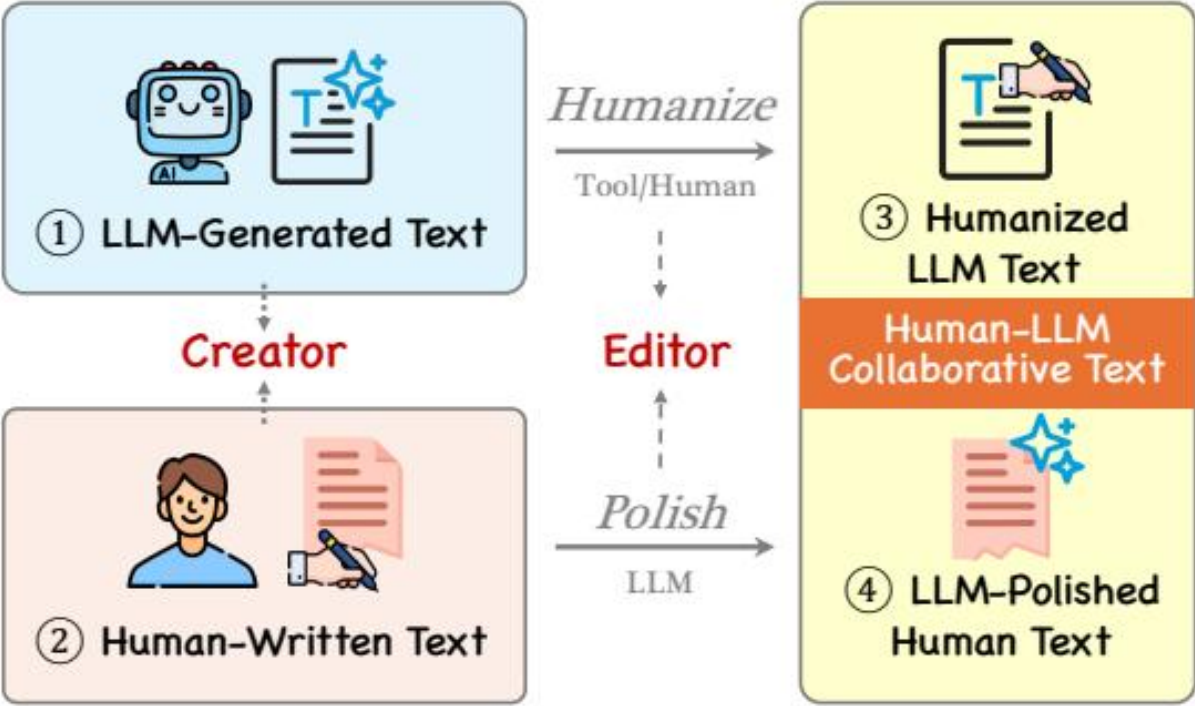


Figure 5: images/e55d24ba4bedda36c670a843a1c72b2aa6686087905fe1e28eca983d66b17f36.jpg

(b)

(b)

Most Existing Works	2-class	① LLM	② Human	
	3-class	① LLM	② Human	③+④ Collaborative
Our Focus	4-class	① LLM	② Human	③ Humanized ④ Polished

Figure 6: images/0fb2e4696fa12a34638d7e9fd5e5c32f1f294c1ccd889e3d855b2abf75bfcc52.jpg

Figure 1: Illustration of our research scope. (a) A Creator-Editor framework for categorizing different types of texts in fine-grained LLM-generated text detection. (b) Comparison of the existing settings and the complex 4-class setting that we focus on in this paper.

图 1：我们的研究范围示意图。(a) 一种创作者-编辑框架，用于在细粒度 LLM 生成文本检测中对不同类型文本进行分类。(b) 现有设置与我们本文重点关注的复杂四类设置的对比。

collaboration (Wang et al., 2025). For instance, people may ask LLMs to polish their original drafts for better readability (Yang et al., 2024a); or conversely, humanize LLM-generated outputs to evade detection (Masrour et al., 2025). Such collaborative processes yield hybrid texts that blend the characteristics of human and LLM generations, ultimately blurring the decision boundaries of conventional binary classifiers.

协作 (Wang 等, 2025)。例如, 人们可能会要求大型语言模型 (LLMs) 润色他们的原始草稿以提高可读性 (Yang 等, 2024a); 或者反过来, 使 LLM 生成的输出更具人性化以逃避检测 (Masrour 等, 2025)。这种协作过程会产生混合文本, 融合人类和 LLM 生成文本的特征, 最终模糊传统二元分类器的决策边界。

To address these complexities, recent studies shift towards fine-grained detection settings, typically by introducing a third “mixed” category (Zhang et al., 2024, Artemova et al., 2025, Saha and Feizi, 2025). Yet, even this ternary classification remains insufficient for nuanced LLM-use regulatory policies in specific domains like academic writing (Cahill et al., 2025). Under such policies, text polishing is often considered legitimate writing assistance that requires no compulsory disclosure, while text humanizing for bypassing detectors is often prohibited, as it brings improper advantages to cheating students and damages academic integrity.

为了解决这些复杂性, 最近的研究转向更细粒度的检测设置, 通常通过引入一个第三类“混合”类别 (Zhang 等, 2024; Artemova 等, 2025; Saha 和 Feizi, 2025)。然而, 即使这种三分类方法在特定领域 (如学术写作) 中仍不足以满足细致的 LLM 使用监管政策 (Cahill 等, 2025)。在这些政策下, 文本润色通常被视为合法的写作辅助手段, 无需强制披露; 而为了绕过检测器进行的人类化文本则通常被禁止, 因为它会给作弊学生带来不当的优势, 损害学术诚信。

In this paper, we study a practically important four-class detection setting where the mixed category is explicitly separated into LLM-Polished Human Text and Humanized LLM Text classes. Inspired by the conceptual framework from Bao et al. (2025) and prior four-class fine-grained detection efforts such as DetectAIve (Abassy et al., 2024), we analyze the four classes through the dual lenses of creator and editor and propose to enhance the modeling of creators’ contributions for fine-grained detection. As illustrated in Figure 1, the creator establishes the basic elements and logical flow, while the editor controls the linguistic expression and surface-level style of these elements. For the pure human/LLM classes, differentiating the two roles is unnecessary; thus, conventional binary classifiers only need to obtain unified features to model human-LLM differences. In contrast, the creator-editor collaboration modes for the two mixed classes are quite different: LLM-Polished Human Text originates from a human creator’s framework and is subsequently refined by an LLM’s stylistic surface, whereas Humanized LLM Text has an LLM-generated foundation but is then edited by humans to perturb LLM traits. These divergent modes produce unique traits that are hard for unified features to capture, making it essential to look beyond the final actor and model the contributions of the creator and editor roles separately.

在本文中, 我们研究了一个具有实际意义的四类检测设置, 其中混合类别被明确地分为 LLM 优化的人类文本和人类化 LLM 文本类别。受 Bao 等人 (2025) 的概念框架以及之前如 DetectAIve (Abassy 等人, 2024) 等四类细粒度检测工作的启发, 我们通过创作者和编辑的双重视角分析这四个类别, 并提出增强对创作者贡献的建模以实现细粒度检测。如图 1 所示, 创作者建立基本元素和逻辑流程, 而编辑则控制这些元素的语言表达和表面风格。对于纯人类/LLM 类别, 区分这两个角色是不必要的; 因此, 传统的二分类器只需获取统一特征即可建模人类与 LLM 之间的差异。相比之下, 这两个混合类别的创作者-编辑协作模式差异很大: LLM 优化的人类文本来源于人类创作者的框架, 并随后由 LLM 进行风格层面的润色; 而人类化 LLM 文本则具有 LLM 生成的基础, 但随后由人类进行编辑以扰动 LLM 的特征。这些不同的模式会产生独特的特征, 这些特征难以通过统一特征捕捉, 因此有必要超越最终执行者, 分别建模创作者和编辑角色的贡献。

To address the four-class detection challenge, we propose the Rhetorical Analysis for Creator-Editor Modeling (RACE) that explicitly models the distinct contributions of the creator and the editor. RACE is grounded in the argument that an editor’s influence is primarily manifested in the linguistic expression, while the creator’s identity is deeply rooted in the logical organization and argumentative progression of the content. To model the editor’s role, RACE first segments the text into Elementary Discourse Units (EDUs) and extracts their semantic representations, which reflect surface-level linguistic choices and refinements. To model the creator’s role, RACE utilizes Rhetorical Structure Theory (RST) (Mann and Thompson, 1988) to construct an EDU-based logical relation graph that characterizes the foundational organization of the text, highlighting the human-LLM creation differences stemming from their fundamental knowledge formation mechanisms. The graph is then processed by rhetoric-guided message passing to propagate information to capture complex rhetorical dependencies, which produces a root pooling representation

for final prediction. Our contributions are as follows:

为了解决四类检测挑战，我们提出了面向创作者-编辑建模的修辞分析（RACE），该方法明确建模了创作者和编辑的各自独特贡献。RACE 基于一个论点：编辑的影响主要体现在语言表达上，而创作者的身份则深深植根于内容的逻辑组织和论证进展中。为了建模编辑的角色，RACE 首先将文本分割为基本话语单元 (EDUs)，并提取其语义表示，这些表示反映了表层的语言选择和润色。为了建模创作者的角色，RACE 利用修辞结构理论 (RST) (Mann and Thompson, 1988) 构建一个基于 EDU 的逻辑关系图，该图描述了文本的基础组织结构，突出了人类与 LLM 创作差异，这些差异源于其基本知识形成机制。然后，通过修辞引导的信息传递处理该图，以传播信息并捕捉复杂的修辞依赖关系，从而生成一个根池化表示用于最终预测。我们的贡献如下：

- **Task Specificity:** We introduce a task-specific detection framework for the four-class setting that explicitly identifies LLM-Polished Human Text and Humanized LLM Text.
- **Method:** We propose RACE, a creator-editor modeling approach that leverages Rhetorical Structure Theory to capture the creator’s logical organization and the editor’s linguistic refinements for fine-grained detection.
- **Performance:** Extensive experiments demonstrate the superiority of RACE under the four-class fine-grained setting with low false alarms.
- **任务特定性：** 我们引入了一个针对四类设置的任务特定检测框架，明确识别出经过大语言模型润色的人类文本和人类化的大语言模型文本。
- **方法：** 我们提出了 RACE，一种创作者-编辑建模方法，利用修辞结构理论来捕捉创作者的逻辑组织和编辑者的语言润色，以实现细粒度检测。
- **性能：** 大量实验表明，在四类细粒度设置下，RACE 具有优越的性能，并且误报率较低。

Reference	Target	Mean	Std
Human-Written	LLM-Polished	0.92	0.08
Human-Written	LLM-Generated	0.84	0.12
Human-Written	Humanized	0.86	0.12
LLM-Generated	Human-Written	0.84	0.12
LLM-Generated	LLM-Polished	0.89	0.09
LLM-Generated	Humanized	0.95	0.07

参考	目标	平均值	Std
人工撰写	LLM-Polished	0.92	0.08
人工撰写	LLM 生成	0.84	0.12
人工撰写	人性化	0.86	0.12
LLM 生成	人工撰写	0.84	0.12
LLM 生成	LLM-Polished	0.89	0.09
LLM 生成	人性化	0.95	0.07

Table 1: Sample-level cosine similarity between rhetorical-relation frequency vectors on HART dataset.

表 1: HART 数据集上修辞关系频率向量的样本级余弦相似度。

2. Preliminaries

2. 前言

2.1. Rhetorical Structure Theory

2.1. 修辞结构理论

Rhetorical Structure Theory (RST) is a descriptive framework for natural text organization, originally proposed to analyze how coherent discourse is constructed (Mann and Thompson, 1988). RST models the hierarchical and functional dependencies between text spans, treating a text piece not as a linear sequence of words but as a structured tree of logical intentions.

修辞结构理论 (RST) 是一个描述性框架, 用于分析自然文本的组织方式, 最初提出是为了分析连贯话语是如何构建的 (Mann 和 Thompson, 1988)。RST 模型描述了文本片段之间的层次结构和功能依赖关系, 将文本视为一系列逻辑意图的结构树, 而不是简单的线性词序。

The construction of a structured tree begins with segmenting the text into several spans called Elementary Discourse Units, which are typically clauses or phrases. Then the text spans are linked through rhetorical relations (e.g., Elaboration, Contrast, Cause), resulting in an RST tree. The RST tree can serve as a fingerprint of the creator's thought process. Specifically, we posit that human and machine creators exhibit distinct structural signatures. For humans, the writing process is inherently teleological, employing complex

构建一个结构化的树开始于将文本分割成若干个称为基本话语单元 (Elementary Discourse Units) 的片段, 这些单元通常是句子或短语。然后通过修辞关系 (例如: 扩展、对比、因果) 将文本片段连接起来, 从而形成一个 RST 树。RST 树可以作为创作者思维过程的指纹。具体来说, 我们认为人类和机器创作者表现出不同的结构特征。对于人类而言, 写作过程本质上是目的性的, 采用复杂的

rhetorical relations such as clausal coordination to guide readers through a preset logical progression. In contrast, LLMs, driven by auto-regressive probability, prioritize informational density over narrative logic, resulting in superficial structural signatures (Reinhart et al., 2025). By modeling logical organization, we can capture the intrinsic differences in how humans and machines architect their narratives.

诸如从句协调等修辞关系, 用于引导读者沿着预设的逻辑进程前进。相比之下, 大型语言模型 (LLMs) 受自回归概率驱动, 更注重信息密度而非叙事逻辑, 导致表面的结构特征 (Reinhart 等, 2025)。通过建模逻辑组织, 我们可以捕捉人类与机器在构建叙事时内在差异。

2.2. Motivating Analysis

2.2. 动机分析

We conducted a preliminary statistical analysis on the distribution of RST relations across the HART dataset (Bao et al., 2025). Specifically, we adopt the Z-score to measure the deviation of each relation's frequency. For an RST relation j in class k , the Z-score is calculated as $Z_{k,j} = (\bar{x}_{k,j} - \mu_j) / \sigma_j$, where $\bar{x}_{k,j}$ is the intra-class mean of relative frequency, and μ_j, σ_j are global mean and standard deviation. A value of $Z > 0$ indicates over-expression relative to the general population.

我们对 HART 数据集 (Bao 等, 2025) 中 RST 关系的分布进行了初步的统计分析。具体来说, 我们采用 Z-score 来衡量每个关系频率的偏离程度。对于类别 k 中的 RST 关系 j , 其 Z-score 计算公式为

$Z_{k,j} = (\bar{x}_{k,j} - \mu_j) / \sigma_j$ ，其中 $\bar{x}_{k,j}$ 是类别内相对频率的均值， μ_j, σ_j 是全局均值和标准差。一个值为 $Z > 0$ 的结果表示相对于一般人群而言存在过度表达。

As visualized in Figure 2 (a), human creators show a significant over-expression in Attribution and Temporal, indicating a discourse structure that is better grounded in external sources and more contextually organized. LLM creators, conversely, exhibit strong spikes in Elaboration and Cause, suggesting a stronger reliance on local expansion, lacking the deep intertextual grounding found in human writing. In Figure 2 (b), even after

如图 2(a) 所示，人类创作者在归属和时间维度上表现出显著的过度表达，表明其话语结构更基于外部来源，并且在上下文中更为组织有序。相比之下，LLM 创作者在扩展和因果维度上表现出强烈的峰值，这表明其更依赖于本地扩展，缺乏人类写作中发现的深层次的互文性基础。在图 2(b) 中，即使在.....

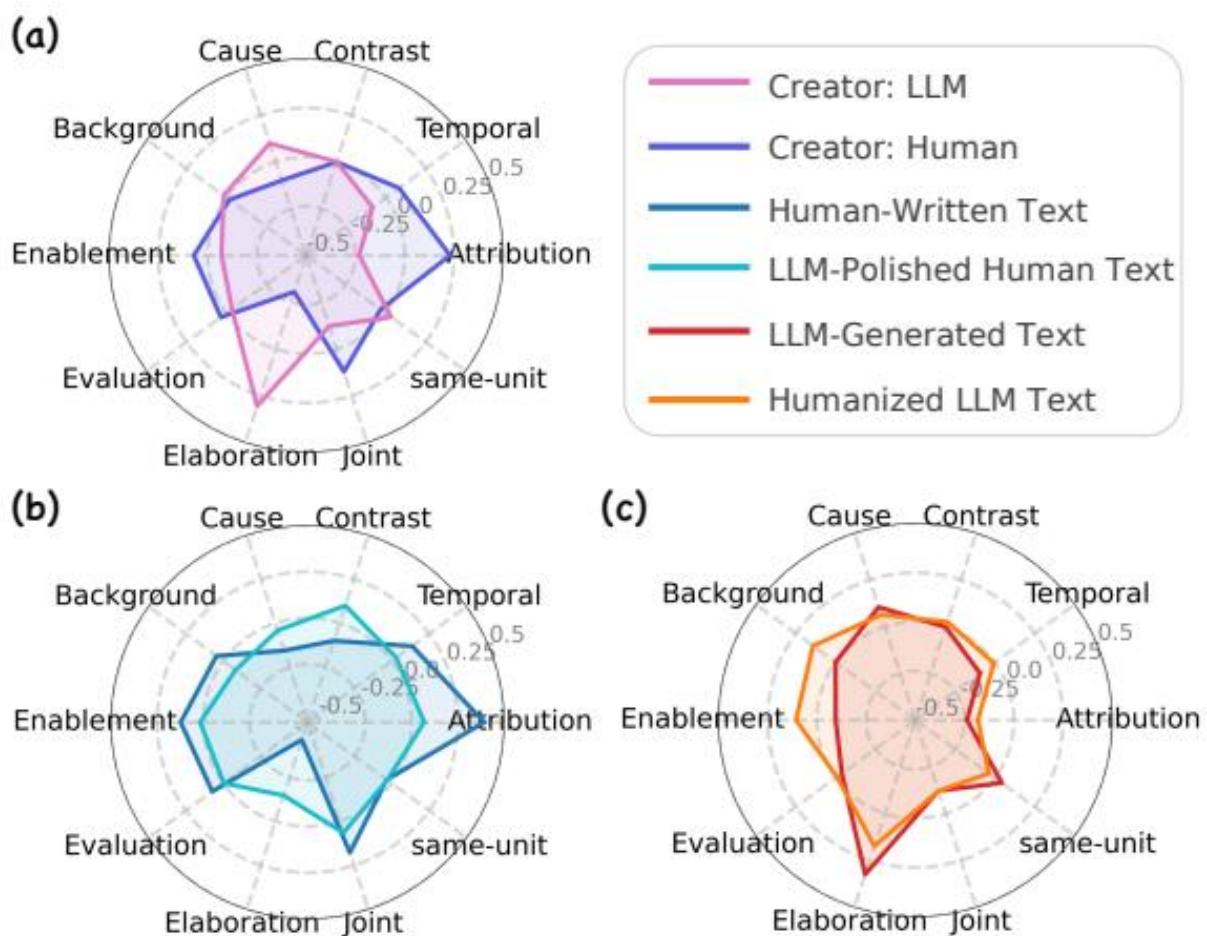


Figure 2: Distribution of RST relations. (a) Divergence of Creators: Human creators build deeper rhetorical hierarchies (e.g., Attribution, Temporal), whereas LLMs produce flatter structures relying on surface-level relations (e.g., Elaboration, Cause). (b) LLM-Polished: underlying human architecture persists. (c) Humanized: underlying LLM architecture persists.

图 2：RST 关系的分布。(a) 创作者的分歧：人类创作者构建更深层次的修辞层次结构（例如，归属、时间），而 LLMs 则依赖于表层关系构建较为扁平的结构（例如，扩展、因果）。(b) LLM 优化：底层的人类架构得以保留。(c) 人性化：底层的 LLM 架构得以保留。

LLMs' polishing, the text retains the high Attribution features, which are more aligned with the human creator. Similarly, Figure 2 (c) shows that human editing fails to mask the underlying LLMs' structural signatures, as Elaboration remain dominant.

LLMs 的润色，文本保留了高归因特征，这些特征更符合人类创作者。同样，图 2(c) 显示，人类编辑无法掩盖底层 LLMs 的结构特征，因为详述仍占主导地位。

Using each document’s rhetorical-relation frequency vector (e.g., {‘Elaboration’: 5, ‘Contrast’: 2} $\square$ [5, 2]), we compute pairwise cosine similarity between a source text and its variants to give a complementary sample-level analysis. Table 1 indicates that texts produced by the same creator remain rhetorically closer to each other even after subsequent editing, further supporting the creator-editor hypothesis.

使用每个文档的修辞关系频率向量（例如，{‘Elaboration’: 5, ‘Contrast’: 2} $\square$ [5, 2]），我们计算源文本与其变体之间的成对余弦相似度，以提供互补的样本级分析。表 1 表明，即使经过后续编辑，同一创作者生成的文本在修辞上仍然彼此更为接近，进一步支持了创作者-编辑假设。

These findings indicate that the subsequent editing operation generally preserves the underlying logic of the creator, which shows the possibility of separately modeling the unique characteristics of humans and LLMs as creators or editors. In the next section, we will introduce rhetorical structure information to model the dual roles for fine-grained LLM-generated text detection.

这些发现表明，后续的编辑操作通常保留了创作者的底层逻辑，这表明可以分别建模人类和大型语言模型作为创作者或编辑的独特特征。在下一节中，我们将引入修辞结构信息，以建模双重角色，用于精细粒度的大型语言模型生成文本检测。

3. Proposed Method: RACE

3. 提出的方法：RACE

To capture the dual trace of creator and editor for fine-grained LLM-generated text detection, we propose the logical-structure-aware detection framework, RACE. As illustrated in Figure 3, RACE consists of four key components: Dual Trace Extraction, Logic-Aware Graph Initialization, Rhetoric-Guided Message Passing, and Graph Readout and Classification. Through these modules, RACE models the generative process through the dual lenses of linguistic expression and logical organization to improve fine-grained detection performance.

为了捕捉创作者和编辑的双重痕迹，以实现对细粒度生成文本的检测，我们提出了一个基于逻辑结构的检测框架 RACE。如图 3 所示，RACE 包含四个关键组件：双重痕迹提取、逻辑感知图初始化、修辞引导的消息传递以及图读取与分类。通过这些模块，RACE 从语言表达和逻辑组织两个角度对生成过程进行建模，以提高细粒度检测性能。

3.1. Dual Trace Extraction

3.1. 双踪提取

To transform unstructured raw text into a structured logic-aware representation, we utilize the end-to-end RST parser developed by Chistova (2024), which achieves superior performance in identifying hierarchical discourse dependencies.

为了将非结构化的原始文本转换为具有结构化逻辑表示的文本，我们使用了 Chistova (2024) 开发的端到端 RST 解析器，该解析器在识别层次化语篇依赖关系方面表现出色。

Formally, the parsing process is defined as a mapping function $F_{parse} : D \rightarrow T$. Given an input text piece D , the parser outputs a binary constituency tree T that explicitly encodes the relation topology. In this structure, the leaf nodes constitute the sequence of EDUs $V_{edu} = \{u_1, u_2, \dots, u_{|V_{edu}|}\}$, where each u_i aligns with a specific continuous text span $[s_i, e_i]$. Recursively, the internal nodes $V_{rel} = \{v_1, v_2, \dots, v_{|V_{rel}|}\}$ capture the logical organization by assigning a specific rhetorical label $r \in R$ (e.g., Elaboration, Contrast) to the dependencies between sub-trees. The tree T serves as the foundational skeleton, which is subsequently transformed into a logic-aware multi-relational graph to enable rhetoric-guided message passing.

正式来说，解析过程被定义为一个映射函数 $F_{parse} : D \rightarrow T$ 。给定一个输入文本片段 D ，解析器输出一个二叉成分树 T ，该树显式地编码了关系拓扑。在这个结构中，叶节点构成了 EDUs $V_{edu} = \{u_1, u_2, \dots, u_{|V_{edu}|}\}$ 的序列，其中每个 u_i 对应于一个特定的连续文本片段 $[s_i, e_i]$ 。递归地，内部节点 $V_{rel} = \{v_1, v_2, \dots, v_{|V_{rel}|}\}$ 通过为子树之间的依赖关系分配一个特定的修辞标签 $r \in R$ （例如：扩展、对比）来捕捉逻辑组织。树 T 作为基础骨架，随后被转换为一个具有逻辑感知的多关系图，以实现修辞引导的消息传递。

3.2. Logic-Aware Graph Initialization

3.2. 基于逻辑的图初始化

Building upon the parsed tree T , the text piece is formalized as a multi-relational graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, where $V = V_{edu} \cup V_{rel}$. Each edge $e \in E$ is represented as a triplet (u, r, v) , preserving the explicit dependency structure where relation nodes govern their constituent EDUs.

在解析树 T 的基础上，文本片段被形式化为一个多关系图 $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ ，其中 $V = V_{edu} \cup V_{rel}$ 。每条边 $e \in E$ 被表示为一个三元组 (u, r, v) ，保留了显式的依赖结构，其中关系节点控制其构成的 EDUs。

Furthermore, a hybrid strategy combining descendant span pooling with information bottleneck projection is proposed to initialize non-leaf nodes with semantically-informed representations, thus going beyond surface-level relation labels to encode richer contextual information.

此外，提出了一种结合后代跨度池化与信息瓶颈投影的混合策略，用于用语义引导的表示初始化非叶节点，从而超越表面关系标签，编码更丰富的上下文信息。

Descendant Span Pooling. For a text piece D with tokens $\{t_1, \dots, t_K\}$, a pre-trained language model (PLM) is employed as the backbone to produce a sequence of contextualized embeddings $E \in R^{K \times d_{PLM}}$. The content

后代跨度池化。 对于一个文本片段 D ，其包含的 token 为 $\{t_1, \dots, t_K\}$ ，采用一个预训练的语言模型 (PLM) 作为骨干网络，生成一个上下文化的嵌入序列 $E \in R^{K \times d_{PLM}}$ 。内容

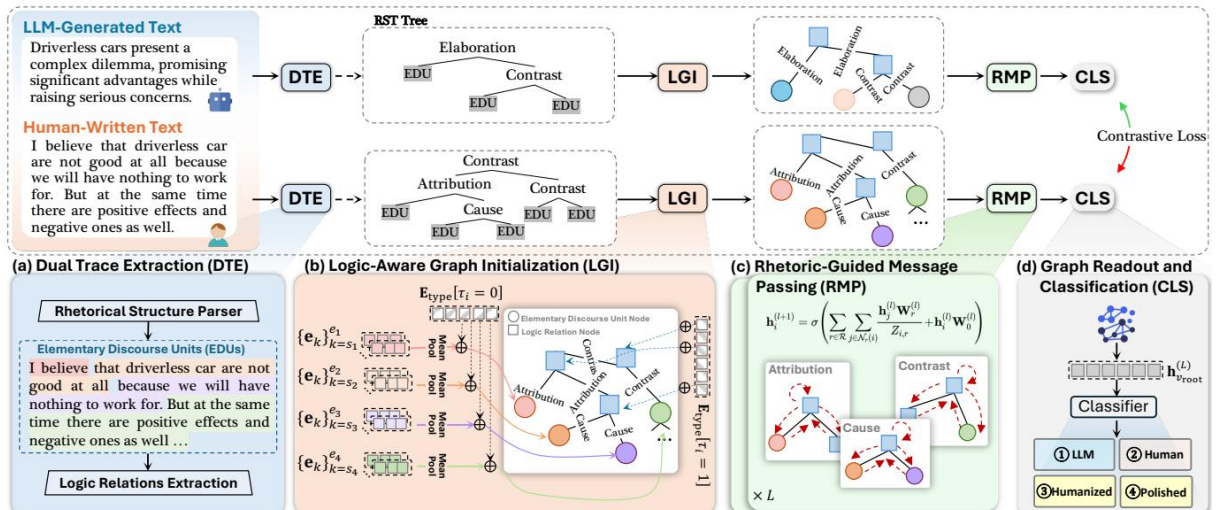


Figure 7: images/f15762c087e2dfd620940f158102865ea8eb31e9c43cd4fc283968c19d86c704.jpg

Figure 3: Overall architecture of RACE. Given a text piece, RACE (a) first captures both creator and editor traces through rhetorical structure construction and elementary discourse unit extrac-

tion. (b) These dual traces are then transformed into a logic-aware graph, where both linguistic expression and logical organization signals are encoded into node features via descendant span pooling and relation-aware projection. (c) Next, Rhetoric-Guided Message Passing propagates information through relation-specific aggregation with basis decomposition to capture complex rhetorical dependencies. (d) Finally, the global text representation is obtained via root pooling for classification.

图 3: RACE 的整体架构。给定一段文本, RACE (a) 首先通过修辞结构构建和基本话语单元提取来捕捉创作者和编辑的痕迹。(b) 然后, 这些双重痕迹被转换为一个具有逻辑感知的图, 其中语言表达和逻辑组织信号通过子跨度池化和关系感知投影编码到节点特征中。(c) 接下来, 修辞引导的消息传递通过基分解进行关系特定的聚合, 以捕捉复杂的修辞依赖关系。(d) 最后, 通过根池化获得全局文本表示以进行分类。

representation $\mathbf{c}_i$ for any node $v_i \in V$ is computed recursively:

节点 $v_i \in V$ 的表示 $\mathbf{c}_i$ 是递归计算的:

$$\mathbf{c}_i = \begin{cases} \text{MeanPool} \left(\{\mathbf{e}_k\}_{k=s_i}^{e_i} \right), & \text{if } v_i \in \mathcal{V}_{\text{edu}} \\ \frac{1}{|\mathcal{D}(v_i)|} \sum_{u \in \mathcal{D}(v_i)} \mathbf{c}_u, & \text{if } v_i \in \mathcal{V}_{\text{rel}}, \end{cases} \quad (1)$$

where e_k is the k -th row of $\mathbf{E}$, and $\mathcal{D}(v_i) \subset \mathcal{V}_{\text{edu}}$ denotes the set of all descendant nodes in the sub-tree rooted at v_i . This strategy ensures that relation nodes are initialized with the global semantic centroid of the text segments they govern.

其中 e_k 是 $\mathbf{E}$ 的第 k 行, $\mathcal{D}(v_i) \subset \mathcal{V}_{\text{edu}}$ 表示以 v_i 为根的子树中所有后代节点的集合。这种策略确保关系节点被初始化为它们所支配的文本片段的全局语义中心。

Information Bottleneck Projection. Raw semantic embeddings often contain surface-level lexical noise irrelevant to structural authorship analysis. To filter this redundancy, a dimension reduction strategy is adopted as an information bottleneck. Specifically, the PLM embeddings are projected into a compact structural space of dimension d_{feat} :

信息瓶颈投影。原始的语义嵌入通常包含与结构化作者身份分析无关的表层词汇噪声。为了过滤这种冗余, 采用了一种降维策略作为信息瓶颈。具体而言, 将 PLM 嵌入投影到一个紧凑的结构空间中, 维度为 d_{feat} :

$$\begin{aligned} \mathbf{c}'_i &= \mathbf{c}_i + \mathbf{E}_{\text{type}}[\tau_i], \\ \mathbf{h}_i^{(0)} &= \text{Dropout} \left(\text{LN} \left(\mathbf{c}'_i \mathbf{W}_{\text{proj}} + \mathbf{b}_{\text{proj}} \right) \right), \end{aligned} \quad (2)$$

where $\tau_i \in \{0, 1\}$ indicates the node type (EDU or Relation), $\mathbf{E}_{\text{type}}$ is the learnable node type embedding table, $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d_{\text{PLM}} \times d_{\text{feat}}}$ and $\mathbf{b}_{\text{proj}}$ are the projection parameters, and LN signifies layer normalization. This compression forces the model to distill only the most salient features required for the subsequent rhetoric-guided message passing.

其中 $\tau_i \in \{0, 1\}$ 表示节点类型 (EDU 或 Relation), $\mathbf{E}_{\text{type}}$ 是可学习的节点类型嵌入表, $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d_{\text{PLM}} \times d_{\text{feat}}}$ 和 $\mathbf{b}_{\text{proj}}$ 是投影参数, LN 表示层归一化。这种压缩迫使模型仅蒸馏后续修辞引导的消息传递所需的最关键特征。

3.3. Rhetoric-Guided Message Passing

3.3. 修辞引导的消息传递

To learn the human-LLM differences over complex rhetorical dependencies, an L-layer Relational Graph Convolutional Network (RGCN; Schlichtkrull et al., 2018) is adopted on the logic-aware graph. Unlike vanilla GCNs that treat all edges uniformly (Kipf and Welling, 2017), RGCN assigns relation-specific transformation matrices, allowing the model to learn distinct propagation rules for different rhetorical logics.

为了学习人类与 LLM 在复杂修辞依赖关系上的差异，采用了一种 L 层的关系图卷积网络（RGCN; Schlichtkrull 等，2018）应用于逻辑感知图上。与将所有边视为相同的普通 GCN（Kipf 和 Welling, 2017）不同，RGCN 为每种关系分配特定的变换矩阵，使得模型能够为不同的修辞逻辑学习不同的传播规则。

Message Aggregation. In each layer l , the node representation is updated as:

信息聚合。在每一层 l 中，节点表示被更新为：

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} \frac{\mathbf{h}_j^{(l)} \mathbf{W}_r^{(l)}}{Z_{i,r}} + \mathbf{h}_i^{(l)} \mathbf{W}_0^{(l)} \right), \quad (3)$$

where $\sigma(\cdot)$ denotes the activation function, $\mathbf{W}_r^{(l)}$ is the relation-specific weight matrix for relation r in the l -th layer, $\mathcal{N}_r(i)$ is the set of neighbors under relation r , $Z_{i,r}$ is a normalization constant, and $\mathbf{W}_0^{(l)}$ handles the self-loop update.

其中 $\sigma(\cdot)$ 表示激活函数， $\mathbf{W}_r^{(l)}$ 是第 l 层中关系 r 的关系特定权重矩阵， $\mathcal{N}_r(i)$ 是关系 r 下的邻居集合， $Z_{i,r}$ 是归一化常数， $\mathbf{W}_0^{(l)}$ 处理自环更新。

Basis Decomposition for Regularization. Given the large number of fine-grained rhetorical relations, learning $\mathbf{W}_r^{(l)}$ for each relation leads to parameter explosion and overfitting. To constrain the weight space, basis decomposition is employed:

正则化的基本分解。鉴于细粒度修辞关系的数量很大，为每个关系学习 $\mathbf{W}_r^{(l)}$ 会导致参数爆炸和过拟合。为了约束权重空间，采用基本分解方法：

$$\mathbf{W}_r^{(l)} = \sum_{k=1}^B \alpha_{rk}^{(l)} \mathbf{V}_k^{(l)}, \quad (4)$$

where $\{\mathbf{V}_k^{(l)}\}_{k=1}^B$ is a set of shared basis matrices, and $\alpha_{rk}^{(l)}$ are learnable scalar coefficients unique to relation r . This technique forces the model to learn the “atomic” components of rhetorical logic, improving generalization on sparse relations.

其中 $\{\mathbf{V}_k^{(l)}\}_{k=1}^B$ 是一组共享的基础矩阵，而 $\alpha_{rk}^{(l)}$ 是关系 r 特有的可学习标量系数。这种技术迫使模型学习修辞逻辑的“原子”组成部分，从而提高在稀疏关系上的泛化能力。

3.4. Graph Readout and Classification

3.4. 图读取与分类

As the logical structure is inherently hierarchical with a single root node v_{root} encompassing the entire text piece’s rhetorical intent, a root pooling strategy is employed to capture the global text representation. The global representation $\mathbf{z}_{\mathcal{G}}$ is directly extracted from the root node’s final hidden state:

由于逻辑结构本质上是具有单一根节点 v_{root} 的层次结构，该节点涵盖了整个文本片段的修辞意图，因此采用根节点池化策略来捕捉全局文本表示。全局表示 $\mathbf{z}_{\mathcal{G}}$ 是直接根节点的最终隐藏状态中提取的：

$$\mathbf{z}_{\mathcal{G}} = \mathbf{h}_{v_{\text{root}}}^{(L)}. \quad (5)$$

Finally, the global representation $\mathbf{z}_{\mathcal{G}}$ is passed to a classification head:

最后，全局表示 $\mathbf{z}_{\mathcal{G}}$ 被传递给一个分类头：

$$\tilde{\mathbf{z}} = \sigma(\text{Dropout}(\mathbf{z}_{\mathcal{G}})\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}),$$

$$\hat{y} = \text{Softmax}(\text{Dropout}(\tilde{\mathbf{z}})\mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}}), \quad (6)$$

where $\mathbf{W}_{\text{in}}, \mathbf{W}_{\text{out}}$ are weight matrices and $\mathbf{b}_{\text{in}}, \mathbf{b}_{\text{out}}$ are bias vectors, σ is a non-linear activation function, and $\hat{y}$ is the predicted probability.

其中 $\mathbf{W}_{\text{in}}$ 和 $\mathbf{W}_{\text{out}}$ 是权重矩阵， $\mathbf{b}_{\text{in}}$ 和 $\mathbf{b}_{\text{out}}$ 是偏置向量， σ 是一个非线性激活函数， $\hat{y}$ 是预测的概率。

Optimization. RACE is optimized using a joint loss that combines the supervised contrastive loss (Khosla et al., 2020) $\mathcal{L}_{\text{con}}$ and the cross-entropy loss $\mathcal{L}_{\text{ce}}$. The former is applied to the normalized feature representations, encouraging the model to learn a compact representation space. The joint loss function is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{con}} + \mathcal{L}_{\text{ce}}$.

优化。RACE 使用一个联合损失函数进行优化，该损失函数结合了监督对比损失 (Khosla 等, 2020) $\mathcal{L}_{\text{con}}$ 和交叉熵损失 $\mathcal{L}_{\text{ce}}$ 。前者应用于归一化的特征表示，鼓励模型学习一个紧凑的表示空间。联合损失函数为 $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{con}} + \mathcal{L}_{\text{ce}}$ 。

4. Experiments

4. 实验

4.1. Experimental Setup

4.1. 实验设置

Dataset. We use the HART (Bao et al., 2025) benchmark for evaluation due to its coverage of the desired categories. However, the official release only contains validation and test partitions. To enable supervised learning, we reorganized the data distribution (see Appendix A.1) and performed a train/val/test split at the 70:20:10 ratio using stratified sampling across diverse domains (e.g., News, Writing, ArXiv, and Essay), which ensures distribution consistency across all partitions. We further report a stricter group-aware split, where all variants derived from the same base text are assigned to the same partition, in Appendix B.2.

数据集。我们使用 HART (Bao 等, 2025) 基准进行评估, 因为它涵盖了所需的类别。然而, 官方发布的版本仅包含验证集和测试集。为了实现监督学习, 我们重新组织了数据分布 (参见附录 A.1), 并在跨多样领域 (如新闻、写作、ArXiv 和作文) 进行分层抽样的基础上, 按 70:20:10 的比例对数据进行了训练/验证/测试划分, 以确保所有分区的分布一致性。我们进一步在附录 B.2 中报告了一个更严格的组感知划分, 其中所有源自同一基础文本的变体都被分配到同一个分区。

Metrics. To evaluate the quality of the model’s probability estimates independent of arbitrary decision thresholds, we prioritize metrics that assess the global ranking capability of the classifier rather than hard predictions. Specifically, we adopt:

指标。为了评估模型概率估计的质量, 而不受任意决策阈值的影响, 我们优先选择能够评估分类器全局排序能力的指标, 而不是硬预测。具体来说, 我们采用:

- Macro-Averaged AUROC, which evaluates the probability that a randomly selected positive instance from any class is ranked higher than a randomly selected negative instance.
- TPR@1% FPR (True Positive Rate at the 1% False Positive Rate), which requires the detector to make precise judgments while avoiding false alarms (Tufts et al., 2025).
- 宏平均 AUROC, 评估从任何类别中随机选择的正样本被排序在随机选择的负样本之上的概率。
- TPR@1% FPR (在 1% 误报率下的真正例率), 要求检测器在避免误报的同时做出精确判断 (Tufts 等, 2025)。

Baselines. Since there is no task-specific method specifically designed for this 4-class setting, we establish baselines by adapting methods originally designed for two-/three-class settings. For a reasonable comparison, we cover 12 learning-based or metric-based methods and tailor them to the fine-grained setting (More details in Appendix A.3):

基线。由于目前尚无专门针对该 4 类设置的特定任务方法, 我们通过调整最初为二分类或三分类设置设计的方法来建立基线。为了进行合理的比较, 我们涵盖了 12 种基于学习或基于度量的方法, 并将其调整到细粒度设置中 (更多细节请参见附录 A.3):

For learning-based methods, we include RoBERTa (Solaiman et al., 2019), RoBERTa-DANN (Abassy et al., 2024), CoCo (Liu et al., 2023), DeTeCtive (Guo et al., 2024), and LF-Motifs (Kim et al., 2024). For all methods, we increase the number of entries in the classification head (i.e., the fully connected layer) from 2 to 4. CoCo and LF-Motifs incorporate discourse information, while DeTeCtive adopts contrastive learning, which is similar to our method in design.

对于基于学习的方法, 我们包括 RoBERTa (Solaiman 等人, 2019), RoBERTa-DANN (Abassy 等人, 2024), CoCo (Liu 等人, 2023), DeTeCtive (Guo 等人, 2024), 以及 LF-Motifs (Kim 等人, 2024)。对于所有方法, 我们将分类头 (即全连接层) 中的类别数量从 2 增加到 4。CoCo 和 LF-Motifs 融入了话语信息, 而 DeTeCtive 采用了对比学习, 其设计与我们的方法相似。

For metric-based methods, we include Binoculars (Hans et al., 2024), Fast-DetectGPT (F-DetectGPT; Bao et al., 2024), and TDT (West et al., 2025), which typically produce scalars as predictions and thus are hard to directly extend to multi-class scenarios. Here, we conduct necessary modifications to obtain features via these methods by extracting the last-layer representation and appending a learnable MLP for four-class prediction.

对于基于指标的方法, 我们包括 Binoculars (Hans 等, 2024)、Fast-DetectGPT (F-DetectGPT; Bao 等, 2024) 和 TDT (West 等, 2025), 这些方法通常产生标量作为预测结果, 因此难以直接扩展到多类场景。在这里, 我们通过提取最后一层表示并附加一个可学习的 MLP 来获得特征, 以实现四类预测。

Implementation Details. For RACE, we use RoBERTa-base (Liu et al., 2019) as the backbone and only fine-tune the last layer while keeping the preceding layers frozen. The extracted features are projected to a dimension of 128 to initialize node features. The graph component consists of

an RGCN with $L = 2$ layers, a hidden dimension of 512, and 10 bases for parameter regularization. The temperature τ is 0.07 for supervised contrastive loss. We select the best validation checkpoint for testing. All experiments were conducted on a single NVIDIA RTX 4090 GPU.

实现细节。对于 RACE，我们使用 RoBERTa-base (Liu 等人, 2019) 作为主干网络，并仅微调最后一层，而保持前面的层冻结。提取的特征被投影到 128 维以初始化节点特征。图组件由一个具有 $L=2$ 层、隐藏维度为 512 且参数正则化使用 10 个基的 RGCN 组成。监督对比损失的温度 τ 设置为 0.07。我们在验证集上选择表现最好的检查点进行测试。所有实验均在单块 NVIDIA RTX 4090 GPU 上进行。

4.2. Main Results

4.2. 主要结果

Table 2 presents the quantitative comparison of RACE against baselines. We observe that:

表 2 展示了 RACE 与基线模型的定量比较。我们观察到：

1) RACE achieves the highest average performance in TPR@1%FPR. Specifically, RACE outperforms the best baseline CoCo by 3.36% absolute with a low false alarm rate, indicating the effectiveness of the creator-editor dual modeling framework.

1) RACE 在 TPR@1%FPR 上实现了最高的平均性能。具体来说，RACE 在误报率较低的情况下，优于最佳基线 CoCo 3.36%，这表明创作者-编辑双建模框架的有效性。

Method	AUROC	TPR@1%FPR				
Human-Written	LLM-Polished	LLM-Generated	Humanized	Avg		
RoBERTa (Solaiman et al., 2019)	92.22	99.36	68.06	63.14	70.92	75.37
<i>RoBERTa_{DANN}</i> (Abassy et al., 2024)	96.17	96.88	75.03	48.89	71.78	73.14
CoCo (Liu et al., 2023)	97.67	99.68	75.77	63.93	79.43	79.70
SeqXGPT (Wang et al., 2023)	89.87	98.38	15.23	14.32	31.68	39.90
DeTeCtive (Guo et al., 2024)	95.74	98.62	0.00	0.00	77.23	43.96
LF-Motifs (Kim et al., 2024)	98.20	96.68	69.61	67.01	75.62	77.23
Binoculars (Hans et al., 2024)	79.15	29.49	7.34	4.37	5.50	11.70
<i>Binoculars_{MLP}</i> (Hans et al., 2024)	50.03	0.00	0.00	0.00	0.00	0.00
<i>Binoculars_{CoCo}</i> (Bao et al., 2025)	50.03	0.00	0.00	0.00	0.00	0.00

F-DetectGPT (Bao et al., 2024)	61.70	0.00	3.37	26.27	0.09	7.70
F-DetectGPT <i>MLP</i> (Bao et al., 2024)	73.69	3.12	3.87	29.35	3.96	10.8
F-DetectGPT <i>C-T</i> (Bao et al., 2025)	49.93	0.00	0.00	0.00	0.00	0.00
<i>TDT_{SVC}</i> (West et al., 2025)	57.16	2.88	2.37	3.58	0.50	2.33
RACE (Ours)	<u>97.99</u> _{±0.13}	99.04 _{±0.40}	83.60 _{±1.61}	74.18 _{±0.95}	75.41 _{±1.03}	83.06 _{±0.57}

方法	AUROC	TPR@1%FPR				
人工撰写	LLM-Polished	LLM 生成	人性化	平均值		
RoBERTa (Solaiman 等, 2019)	92.22	99.36	68.06	63.14	70.92	75.37
<i>RoBERTa_{DANN}</i> (Abassy 等, 2024)	96.17	96.88	75.03	48.89	71.78	73.14
CoCo (Liu 等, 2023)	97.67	99.68	75.77	63.93	79.43	79.70
SeqXGPT (Wang et al., 2023)	89.87	98.38	15.23	14.32	31.68	39.90
DeTeCtive (郭等, 2024)	95.74	98.62	0.00	0.00	77.23	43.96
LF-Motifs (Kim 等, 2024)	98.20	96.68	69.61	67.01	75.62	77.23
望远镜 <i>MLP</i> (Hans 等, 2024)	79.15	29.49	7.34	4.37	5.50	11.70
<i>Binoculars_{C-T}</i> (包等, 2025)	50.03	0.00	0.00	0.00	0.00	0.00
F-DetectGPT (Bao et al., 2024)	61.70	0.00	3.37	26.27	0.09	7.70

F-DetectGPT MLP (Bao et al., 2024)	73.69	3.12	3.87	29.35	3.96	10.8
F-DetectGPT $C-T$ (Bao et al., 2025)	49.93	0.00	0.00	0.00	0.00	0.00
TDT_{SVC} (West 等, 2025)	57.16	2.88	2.37	3.58	0.50	2.33
RACE (Ours)	<u>97.99</u> ± 0.13	99.04 ± 0.40	83.60 ± 1.61	74.18 ± 0.95	75.41 ± 1.03	83.06 ± 0.57

Table 2: Quantitative comparison of detection methods under the 4-class setting. For RACE, we report the results across three runs using different seeds in the format of the mean $\pm std$. All values are reported in percentage (%). Bold and underlined values denote the best and second-best performance, respectively.

表 2: 在 4 类设置下的检测方法定量比较。对于 RACE，我们以平均 $\pm std$ 的格式报告了使用不同种子的三次运行结果。所有值均以百分比 (%) 表示。粗体和下划线的值分别表示最佳和次佳性能。

2) RACE outperforms closely related discourse-aware detection methods. Similar to RACE, CoCo and LF-Motifs utilize discourse information: CoCo relies primarily on entity-coherence graphs to model inner-and inter-sentence relations; while LF-Motifs introduces statistical features of RST trees concatenated with Longformer embeddings. Though they outperform other compared baselines, CoCo struggles to capture the local stylistic shift when semantic entities remain unchanged, and LF-Motifs’s statistical features are relatively shallow. In contrast, RACE leverages RGCN for message passing directly over the relational graph, thereby capturing the intrinsic structural topology and logical anomalies that shallow motifs fail to represent.

2) RACE 在与之密切相关的语篇感知检测方法中表现更优。与 RACE 类似，CoCo 和 LF-Motifs 都利用了语篇信息：CoCo 主要依靠实体一致性图来建模句内和句间的关系；而 LF-Motifs 则将 RST 树的统计特征与 Longformer 嵌入进行拼接。尽管它们在其他对比基线模型上表现更好，但 CoCo 在语义实体保持不变时难以捕捉局部风格的转变，而 LF-Motifs 的统计特征相对较为浅显。相比之下，RACE 直接在关系图上利用 RGCN 进行消息传递，从而捕捉到浅显模式无法表示的内在结构拓扑和逻辑异常。

3) Learning-based methods generally outperform metric-based ones for fine-grained classification. Metrics-based methods typically compress information into scalar values, which may be simple and effective for the binary setting, but the loss that such compression leads to also collapses the high-dimensional feature space necessary for the multi-class task. Aligned with the observation from Tufts et al. (2025), we see poor performance for certain detectors, with TPR@1%FPR as low as 0%. Even if we adopt several modifications to preserve more information, their performance still falls behind the learning-based methods, perhaps because the latter could entail the classification knowledge into well-trained parametric networks.

3) 基于学习的方法在细粒度分类任务中通常优于基于度量的方法。基于度量的方法通常将信息压缩为标量值，这在二分类设置中可能简单且有效，但这种压缩所导致的损失也会压缩多分类任务所需的高维特征空间。与 Tufts 等人 (2025) 的观察结果一致，我们发现某些检测器的性能较差，其 TPR@1%FPR 低至 0%。即使我们采用一些修改以保留更多信息，其性能仍落后于基于学习的方法，也许是因为后者能够将分类知识整合到经过良好训练的参数化网络中。

4.3. Ablation Study

4.3. 消融实验

As presented in Table 3, the TPR@1%FPR shows a clear drop when removing the involved components, confirming their individual benefits for improving fine-grained detection performance. We notice the largest performance drop arises at the LLM-generated class, followed by the LLM-Polished class. This is aligned with our intuition: The LLM-generated and polished samples are created by LLMs and humans, respectively, but share the same editor (i.e. the LLM). The degradation of w/o Relation variant indicates that, without logical relations, vanilla GCN fails to capture patterns defining the role of creator and confirms the core advantage of RACE. Without the contrastive learning that enhances the feature differences and the RGCN that explicitly models the dual roles of creator and editor, the detector might mix the characteristics of the two editor-similar types of samples.

如表 3 所示，当移除相关组件时，TPR@1%FPR 明显下降，这证实了这些组件对提升细粒度检测性能的个体优势。我们注意到性能下降最大的是 LLM 生成的类别，其次是 LLM 优化的类别。这与我们的直觉一致：LLM 生成的样本和优化的样本分别由 LLM 和人类创建，但共享相同的编辑器（即 LLM）。没有关系变体的性能下降表明，没有逻辑关系，普通的 GCN 无法捕捉定义创建者角色的模式，并证实了 RACE 的核心优势。如果没有增强特征差异的对比学习以及显式建模创建者和编辑器双重角色的 RGCN，检测器可能会混淆这两种编辑器相似类型的样本的特征。

Method	AUROC	TPR@1%FPR				
Human-Written	LLM-Polished	LLM-Generated	Humanized	Avg		
RACE	97.99 $\pm$ 0.13	99.04 $\pm$ 0.40	83.60 $\pm$ 1.61	74.18 $\pm$ 0.95	75.41 $\pm$ 1.03	83.06 $\pm$ 0.57
w/o CL	97.73 $\pm$ 0.44	98.21 $\pm$ 0.19	78.07 $\pm$ 1.06	69.10 $\pm$ 2.66	73.43 $\pm$ 1.59	79.70 $\pm$ 1.12
w/o Relation	96.78 $\pm$ 0.65	97.42 $\pm$ 0.75	78.24 $\pm$ 2.08	65.35 $\pm$ 9.19	74.92 $\pm$ 1.74	78.98 $\pm$ 2.95
w/o RGCN	97.91 $\pm$ 0.20	98.92 $\pm$ 0.19	82.27 $\pm$ 4.56	65.35 $\pm$ 6.71	74.42 $\pm$ 1.14	80.24 $\pm$ 2.07
w/o Bottleneck	98.07 $\pm$ 0.22	98.54 $\pm$ 0.26	80.82 $\pm$ 2.45	74.68 $\pm$ 0.86	75.74 $\pm$ 0.99	82.45 $\pm$ 0.60
w/o Basis	97.22 $\pm$ 0.74	97.92 $\pm$ 0.26	83.39 $\pm$ 3.20	73.02 $\pm$ 5.46	74.58 $\pm$ 1.14	82.23 $\pm$ 1.74

方法	AUROC	TPR@1%FPR				
人工撰写	LLM-Polished	LLM 生成	人性化	平均值		
RACE	97.99 $\pm$ 0.13	99.04 $\pm$ 0.40	83.60 $\pm$ 1.61	74.18 $\pm$ 0.95	75.41 $\pm$ 1.03	83.06 $\pm$ 0.57
无 CL	97.73 $\pm$ 0.44	98.21 $\pm$ 0.19	78.07 $\pm$ 1.06	69.10 $\pm$ 2.66	73.43 $\pm$ 1.59	79.70 $\pm$ 1.12
无关系	96.78 $\pm$ 0.65	97.42 $\pm$ 0.75	78.24 $\pm$ 2.08	65.35 $\pm$ 9.19	74.92 $\pm$ 1.74	78.98 $\pm$ 2.95
无 RGCN	97.91 $\pm$ 0.20	98.92 $\pm$ 0.19	82.27 $\pm$ 4.56	65.35 $\pm$ 6.71	74.42 $\pm$ 1.14	80.24 $\pm$ 2.07
无瓶颈	98.07 $\pm$ 0.22	98.54 $\pm$ 0.26	80.82 $\pm$ 2.45	74.68 $\pm$ 0.86	75.74 $\pm$ 0.99	82.45 $\pm$ 0.60
无依据	97.22 $\pm$ 0.74	97.92 $\pm$ 0.26	83.39 $\pm$ 3.20	73.02 $\pm$ 5.46	74.58 $\pm$ 1.14	82.23 $\pm$ 1.74

Table 3: Ablation results (mean $\pm$ std) of RACE. The Bottleneck represents the Information Bottleneck Projection in Eq. (2), the Basis represents the Basis Decomposition in Eq. (4), and w/o Relation means removing the relation types on edges and adopting vanilla GCN (Kipf and Welling, 2017).

表 3: RACE 的消融实验结果 (均值 $\pm std$)。Bottleneck 表示式 (2) 中的信息瓶颈投影, Basis 表示式 (4) 中的基分解, w/o Relation 表示移除边上的关系类型并采用原始的 GCN (Kipf 和 Welling, 2017)。

Variant	AUROC	Avg. TPR@1%FPR
RACE (DMRST)	97.64	82.93
RACE (IsaNLP)	97.99	83.06

变体	AUC (曲线下面积)	平均误报率 1% 时的查全率
RACE (DMRST)	97.64	82.93
RACE (IsaNLP)	97.99	83.06

Table 4: Parser-robustness analysis.

表 4: 解析器鲁棒性分析。

Furthermore, the results of w/o Basis and w/o Bottleneck validate the necessity of parameter efficiency and feature compression. Specifically, removing the Basis Decomposition leads to a noticeable increase in performance variance (i.e., high standard deviation). This means that the basis decomposition is an important regularizer because it forces weight sharing between similar relations. This stops over-parameterization and makes sure that optimization stays stable. Meanwhile, the removal of the Information Bottleneck Projection causes a specific performance degradation on the LLM-Polished class. This corroborates that this module effectively prevents the model from overfitting to superficial patterns shared with the LLM editor and forces it to focus on the invariant features indicative of the human creator.

此外, 去除 Basis 和 Bottleneck 的结果验证了参数效率和特征压缩的必要性。具体来说, 去除 Basis 分解会导致性能方差显著增加 (即标准差较高)。这意味着 Basis 分解是一个重要的正则化手段, 因为它迫使相似关系之间共享权重。这可以防止过度参数化, 并确保优化过程的稳定性。同时, 去除信息瓶颈投影会导致在 LLM-优化类别上出现特定的性能下降。这证实了该模块有效防止模型过度拟合与 LLM 编辑器共享的表面模式, 并迫使模型关注那些表明人类创作者的不变特征。

We also experiment with different RST parsers, IsaNLP (Chistova, 2024) and DMRST (Liu et al., 2021), while keeping the downstream architecture fixed. Table 4 shows that RACE remains stable under parser replacement, suggesting that the core improvement comes from creator-editor modeling rather than artifacts of one parser.

我们还尝试了不同的 RST 解析器, 包括 IsaNLP (Chistova, 2024) 和 DMRST (Liu 等人, 2021), 同时保持下游架构不变。表 4 显示, 在更换解析器后, RACE 仍然保持稳定, 这表明核心改进来自于创作者-编辑建模, 而不是某个解析器的特定特性。

4.4. Further Analysis

4.4. 进一步分析

To provide a deeper insight into the effectiveness of our proposed method, we conduct a comprehensive comparison against CoCo and LF-Motifs, which are top-performing in the main experiments.

为了更深入地了解我们所提出方法的有效性, 我们对其与在主要实验中表现优异的 CoCo 和 LF-Motifs 进行了全面比较。

Discriminability of Feature Representations. To quantitatively evaluate the discriminability of the learned representations, we employ two standard clustering validity indices: the Davies-Bouldin Index (DBI) (Davies and Bouldin, 1979) and the Calinski-Harabasz Index (CH) (Caliński and Harabasz, 1974). As shown in Table 5, RACE achieves a lower DBI of 0.8042 than CoCo (0.9286), indicating a better ratio of intra-cluster scatter to inter-cluster separation. Regarding the CH index, RACE’s score is nearly double that of CoCo, implying that explicitly modeling the creator/editor logic leads to a significantly more compact and distinct embedding space. The indices show the superiority of RACE in learning a discriminative feature space for fine-grained detection.

特征表示的可区分性。为了定量评估所学习的表示的可区分性，我们采用两个标准的聚类有效性指标：Davies-Bouldin 指数 (DBI) (Davies 和 Bouldin, 1979) 和 Calinski-Harabasz 指数 (CH) (Caliński 和 Harabasz, 1974)。如表 5 所示，RACE 的 DBI 值为 0.8042，低于 CoCo 的 0.9286，表明其簇内散度与簇间分离度的比率更好。关于 CH 指数，RACE 的得分几乎是 CoCo 的两倍，这意味着显式建模创建者/编辑器逻辑可以显著地使嵌入空间更加紧凑和区分。这些指标表明，RACE 在学习用于细粒度检测的可区分特征空间方面具有优越性。

Metric	CoCo	RACE
Davies-Bouldin Index ($\square$)	0.9286	0.8042
Calinski-Harabasz Index ($\square$)	2289.40	4333.32

指标	CoCo	RACE
Davies-Bouldin 指数 ($\square$)	0.9286	0.8042
Calinski-Harabasz 指数 ($\square$)	2289.40	4333.32

Table 5: Quantitative evaluation of feature discriminability using clustering validity indices.

表 5：使用聚类有效性指标对特征可区分性进行定量评估。

Domain	Method	AUROC	Avg. TPR@1%FPR
Arxiv	CoCo	93.67	55.91
LF-Motifs	94.07	58.64	
RACE	96.61	76.28	
Essay	CoCo	89.84	28.72
LF-Motifs	91.12	47.85	
RACE	95.88	59.73	
News	CoCo	89.35	39.36
LF-Motifs	91.04	35.36	
RACE	92.69	44.30	
Writing	CoCo	83.03	30.85
LF-Motifs	84.73	31.59	
RACE	86.20	30.84	

领域	方法	AUC (Area Under the Curve)	平均误报率 1% 时的召回率
Arxiv	CoCo	93.67	55.91

LF-Motifs	94.07	58.64	
RACE	96.61	76.28	
作文	CoCo	89.84	28.72
LF-Motifs	91.12	47.85	
RACE	95.88	59.73	
新闻	CoCo	89.35	39.36
LF-Motifs	91.04	35.36	
RACE	92.69	44.30	
写作	CoCo	83.03	30.85
LF-Motifs	84.73	31.59	
RACE	86.20	30.84	

Table 6: Performance comparison under the OOD setting. We employ a Leave-One-Domain-Out protocol where the model is trained on three domains and tested on the fourth unseen domain (Column 1).

表 6: 在 OOD 设置下的性能比较。我们采用 Leave-One-Domain-Out 协议，其中模型在三个域上进行训练，并在第四个未见过的域上进行测试（列 1）。

Impact of Text Length Variations. We investigate how input text length affects detection performance. Figure 4 illustrates the TPR@1% FPR across different token length intervals. While both methods perform well on texts longer than 600 tokens, RACE performs better for shorter texts with 200-600 tokens. This advantage suggests that our graph-based approach is more efficient in capturing the nuanced differences introduced by creators and editors, with relatively limited information.

文本长度变化的影响。我们研究输入文本长度对检测性能的影响。图 4 展示了在不同 token 长度区间内 TPR@1% FPR 的变化情况。虽然两种方法在超过 600 个 token 的文本上表现良好，但在 200-600 个 token 的较短文本上，RACE 表现更优。这种优势表明，我们的基于图的方法在捕捉创作者和编辑引入的细微差异方面更加高效，且所需信息相对有限。

Out-of-Distribution (OOD) Testing.

Out-of-Distribution (OOD) 测试。

To assess the robustness of our method across different text genres, we extend the evaluation to a Leave-One-Domain-Out setting on the four domains in HART,

为了评估我们的方法在不同文本类型中的鲁棒性，我们将评估扩展到在 HART 的四个领域上进行 Leave-One-Domain-Out 设置。

including Arxiv, Essay, News, and Writing. Excluding the preserved domain, the samples in the remaining three domains are split into the training and validation sets with a ratio of 9:1. From Table 6, we see that RACE outperforms CoCo and LF-Motifs in most cross-domain scenarios, particularly for structured genres like research articles and essays. For CoCo, the entity distributions are highly domain-dependent, and when transferred to an unseen domain, the learned entity patterns fail to reflect the new inductive bias, leading to performance degradation. Differently, RACE relies on both linguistic expression and logical organization and forms a more comprehensive view, thus enhancing the OOD generalizability.

包括 Arxiv、Essay、News 和 Writing。排除保留的领域后，其余三个领域的样本按照 9:1 的比例划分为

训练集和验证集。从表 6 中可以看出，在大多数跨领域场景中，RACE 在结构化文体（如研究文章和论文）方面优于 CoCo 和 LF-Motifs。对于 CoCo，实体分布高度依赖于领域，当转移到未见过的领域时，学习到的实体模式无法反映新的归纳偏见，导致性能下降。相反，RACE 依赖于语言表达和逻辑组织，并形成更全面的视角，从而增强对分布外（OOD）数据的泛化能力。

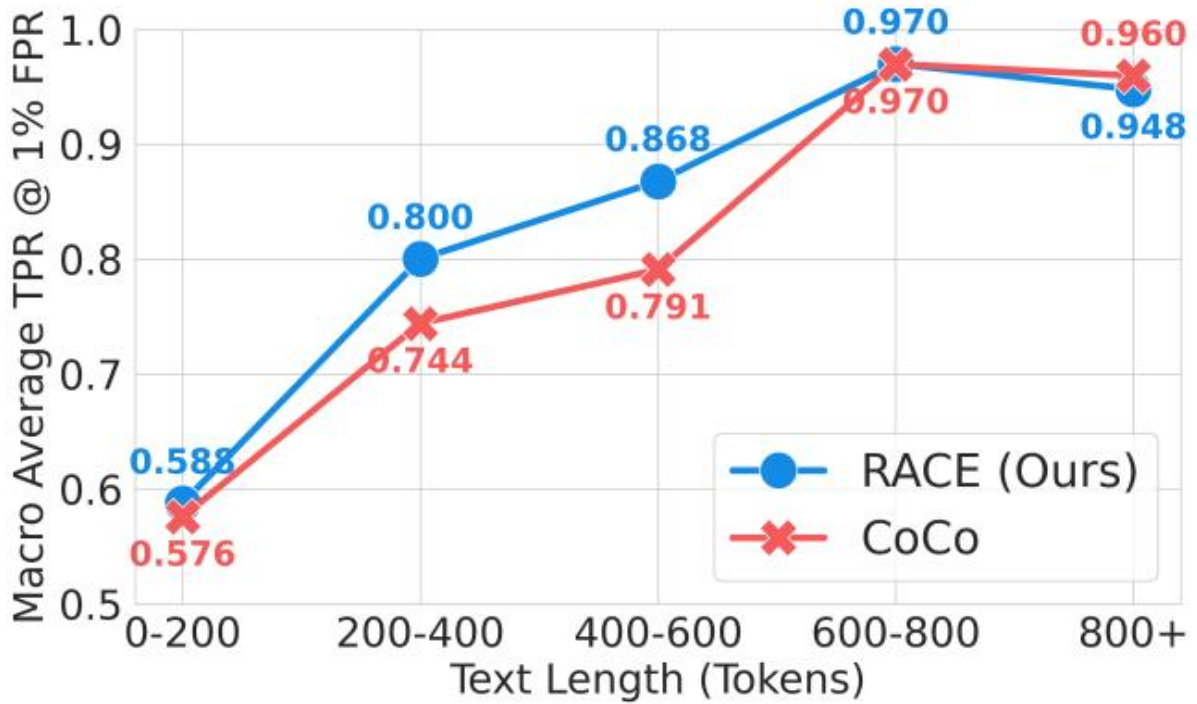


Figure 8: images/01ed936b663d59644e4d758223358536e8f3b8012377687cf2c23bd69d118762.jpg

Figure 4: Analysis of detection performance of CoCo and our proposed RACE across varying text lengths.

图 4: CoCo 和我们提出的 RACE 在不同文本长度下的检测性能分析。